PLANT HEALTH MONITORING SYSTEM USING LEAVES

A PROJECT REPORT

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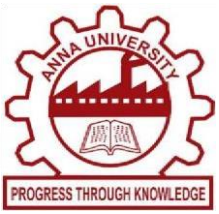
in partial fulfillment for the award of the degree

of

BACHELOR OF ENGINEERING

IN

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ANNA UNIVERSITY REGIONAL CAMPUS, COIMBATORE

ANNA UNIVERSITY : CHENNAI 600 025

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BONAFIDE CERTIFICATE

Certified that this project report **"PLANT HEALTH MONITORING SYSTEM USING LEAVES"** is the bonafide work of "**RANJANADEVI E (710022104024), INDHUMATHI D (710022104028), PRAMOTHINI C (710022104046), SARIKA M (710022104710)"** who carried out the project work under my supervision.

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INTERNAL EXAMINER **EXTERNAL EXAMINER**

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ABSTRACT

The rapid advancement in agriculture and the increasing need for higher crop productivity have made plant disease detection a critical task. Early and accurate identification of plant diseases plays a vital role in reducing crop losses and improving overall yield. This project presents an automated plant health monitoring system using leaf image analysis and deep learning techniques to effectively detect plant diseases and evaluate their severity.

The proposed system analyzes leaf images captured through a camera or uploaded by the user. Initially, the input image undergoes preprocessing techniques such as resizing and normalization. The processed image is then directly fed into a deep learning model, where feature extraction is automatically performed by the network to analyze important characteristics such as color, texture, and patterns for accurate disease detection.

A trained deep learning model is utilized to classify plant diseases with high accuracy. In addition to classification, the system calculates the severity of the disease by estimating the proportion of the affected area on the leaf. A visual heatmap is generated to highlight infected regions, providing better understanding of the results.

Furthermore, the system offers suitable treatment recommendations, including the use of fertilizers and pesticides. A user-friendly graphical interface allows users to upload images, view analysis results, and maintain records for future reference.

The proposed system provides a fast, reliable, and cost-effective solution for plant disease detection and management. This project supports modern agricultural practices and contributes to improved crop health monitoring and sustainable farming.

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LIST OF ABBREVIATIONS

CNN	Convolutional Neural Network
DSS	Decision Support System
GLCM	Gray Level Co-Occurrence Matrix
GRAD-CAM	Gradient-Weighted Class Activation Mapping
GUI	Graphical User Interface
HSV	Hue Saturation Value
JPEG	Joint Photographic Experts Group
JPG	Joint Photographic Experts Group
JSON	Javascript Object Notation
RGB	Red Green Blue
TF	Tensorflow

CHAPTER 1
INTRODUCTION

1.1 OVERVIEW

Agriculture is one of the most important sectors that supports human life and the economy. Maintaining plant health is essential for achieving better crop yield and quality. However, plant diseases can severely affect productivity if not identified at an early stage. Traditional methods of disease detection rely on manual observation, which is time-consuming, less accurate, and requires expert knowledge. Therefore, an automated system is needed to monitor plant health efficiently.

This project, titled “Plant Health Monitoring System Using Leaves,” focuses on detecting plant diseases by analyzing leaf images. Leaves are one of the primary indicators of plant health, as most diseases first appear on them. By using modern image processing and deep learning techniques, the system can identify diseases quickly and accurately.

The proposed system takes leaf images as input, either captured through a camera or uploaded by the user. The images are processed to improve quality and remove noise. Segmentation techniques are applied to separate the leaf from the background, ensuring accurate analysis. Deep learning techniques automatically extract relevant features such as color, texture, and patterns from leaf images for accurate disease detection.

In addition to disease detection, the system provides suggestions for treatment, including suitable fertilizers and pesticides. A simple and user-friendly interface is developed to allow users to upload images, view results, and maintain records.

1.1 PROBLEM STATEMENT

Agriculture is a key sector that contributes to economic growth and food security. Maintaining healthy crops is essential to ensure good yield and quality. However, plant diseases remain a major challenge, as they negatively affect agricultural productivity and crop quality.

In existing practices, disease detection mainly depends on manual observation and the experience of farmers or agricultural experts. These methods are time-consuming and may not always provide accurate results, especially when different diseases show similar symptoms. Moreover, access to expert support is limited in many rural areas, making it difficult for farmers to identify diseases and take appropriate action.

Assessing the severity of plant diseases and selecting suitable treatment methods is also a complex task. Visual inspection alone may not provide reliable results, leading to inefficient use of fertilizers and pesticides and affecting both crop health and environmental sustainability.

In addition, manual monitoring of large agricultural fields requires significant effort and is prone to human error. This reduces the efficiency and consistency of disease detection and management.

To address these challenges, there is a need for an automated and reliable system for plant health monitoring. The proposed Plant Health Monitoring System uses leaf image analysis along with image processing and deep learning techniques to detect plant diseases, estimate their severity, and provide appropriate recommendations. This approach helps in improving crop management and overall agricultural productivity.

1.3 AIM AND OBJECTIVES

The main aim of this project is to develop a Plant Health Monitoring System Using Leaves that can automatically detect plant diseases and assess plant health using leaf image analysis. The system utilizes image processing and deep learning techniques to provide accurate, fast, and reliable results, helping users take timely actions for improving crop productivity.

OBJECTIVES

- To design and develop an automated system for detecting plant diseases using leaf images.
- To apply image processing techniques such as preprocessing and segmentation to extract the leaf region from the background.
- To extract important features like color and texture from leaf images for accurate analysis.
- To implement a deep learning model using Efficientnet Lite0 for classifying different types of plant diseases.
- To estimate the severity of the disease by analyzing the affected area of the leaf.
- To generate a visual representation (heatmap) to highlight infected regions.
- To provide suitable treatment suggestions, including fertilizers and pesticides.
- To develop a user-friendly interface for uploading images and viewing results.
- To maintain records of previous analyses for future reference.
- To create a reliable and cost-effective solution that supports modern agricultural practices.

1.4 SCOPE OF THE PROJECT

The scope of this project is to develop an efficient and automated Plant Health Monitoring System Using Leaf Image Analysis that enables accurate detection of plant diseases and assessment of plant health. The system is designed to assist farmers, agricultural experts, and general users in identifying diseases at an early stage and taking appropriate corrective actions.

The project focuses on analyzing leaf images captured through a camera or uploaded by the user. The input images undergo preprocessing techniques such as resizing, normalization, and noise reduction to improve image quality. The processed images are then fed into a deep learning model based on Convolutional Neural Networks (CNN) for disease classification.

The system utilizes a transfer learning approach using EfficientNet-Lite0, which enhances performance and reduces training time. It is capable of detecting various diseases across multiple plant types. In addition to classification, the system estimates the severity of the disease by calculating the affected area on the leaf using pixel-based analysis.

A visual heatmap representation is generated to highlight infected regions. The system also provides decision support by suggesting appropriate fertilizers and pesticides based on the detected disease.

The application includes a user-friendly graphical interface (GUI) that allows users to upload images, view results, and maintain records.

However, the system is limited to diseases present in the trained dataset and depends on the quality of the input image. Future improvements may include expanding the dataset and enhancing model performance.

CHAPTER 2

LITERATURE REVIEW

2.1 OBJECTIVE OF THE SURVEY

The objective of this literature review is to study existing research related to plant disease detection using leaf image analysis and deep learning techniques. It focuses on understanding various methods used for preprocessing, feature extraction, and classification of plant diseases.

The survey also helps in identifying the limitations of current systems, such as dependency on controlled datasets, lack of real-world adaptability, and absence of advanced features like severity estimation and recommendation systems. This analysis supports the development of an improved and efficient plant health monitoring system.

2.2 MAIN WORKS

Mohanty, Hughes, and Salathé (2016) proposed a CNN-based plant disease detection model using leaf images, achieving high accuracy but with reduced performance under real-world environmental variations such as lighting and background noise [1].

Ferentinos (2018) evaluated deep learning models for plant disease diagnosis and confirmed that CNN architectures are effective for multi-class classification, though they do not address disease severity estimation [2].

Tan and Le (2019) introduced EfficientNet, an efficient convolutional neural network architecture that improves accuracy while reducing computational complexity, and is widely used in image classification applications [3].

Shorten and Khoshgoftaar (2019) discussed image data augmentation techniques that improve model performance by increasing dataset diversity and handling variations in image conditions [4].

Selvaraju, Cogswell, Das, Vedantam, Parikh, and Batra (2017) proposed Grad-CAM, a visualization technique that enhances interpretability by highlighting regions influencing model predictions [5].

Too, Yujian, Njuki, and Yingchun (2019) demonstrated that transfer learning significantly improves the performance of deep learning models for plant disease classification compared to training from scratch [6].

Sladojevic, Arsenovic, Anderla, Culibrk, and Stefanovic (2016) developed a deep learning-based system for plant disease recognition using leaf images, showing strong effectiveness in automated classification tasks [7].

Reddy, Kumar, and Reddy (2019) implemented a CNN-based plant leaf disease detection system, achieving good accuracy while highlighting challenges in real-time deployment and robustness in practical applications [8].

2.3 SUMMARY OF SURVEY

The literature review shows that deep learning techniques, particularly Convolutional Neural Networks (CNNs) and EfficientNet models, are commonly used for plant disease detection and provide good accuracy. Image preprocessing and data augmentation methods help improve model performance, while techniques like Grad-CAM support better understanding of model predictions.

However, some important gaps are identified. Most existing systems focus only on disease classification and do not include disease severity estimation. Many models depend on controlled datasets and may not perform well in real-world conditions. In addition, there is a lack of systems that provide proper treatment recommendations and user-friendly interfaces for practical use.

CHAPTER 3

SYSTEM ANALYSIS

3.1. EXISTING SYSTEM

The existing systems for plant disease detection are predominantly developed based on methodologies discussed in the literature survey, with a primary focus on deep learning and image analysis techniques. Convolutional Neural Networks (CNNs) have been extensively utilized for automated classification of plant diseases using leaf images. These models are trained on large-scale annotated datasets, enabling them to learn discriminative features and achieve high classification accuracy under controlled conditions.

Recent advancements have incorporated architectures such as EfficientNet, which provide an effective balance between accuracy and computational efficiency. Image preprocessing methods including normalization, resizing, and noise reduction are applied to enhance input quality, while data augmentation techniques such as rotation, scaling, and brightness adjustment improve model generalization. Interpretability methods like Grad-CAM are also used to highlight important regions in leaf images that influence predictions.

From an implementation perspective, these systems are deployed through mobile applications, cloud-based platforms, and traditional machine learning models. Mobile applications provide user-friendly interfaces for quick predictions, while cloud-based systems offer higher computational capabilities. Traditional machine learning approaches rely on handcrafted features but are less effective in real-world conditions. Remote sensing techniques are used for large-scale monitoring but are not suitable for detailed leaf-level analysis.

Despite these advancements, existing systems exhibit several limitations. Most models are limited to disease classification and do not include severity assessment. Their dependency on controlled datasets reduces performance in

real-time environments with varying conditions. Additionally, the lack of integrated recommendation systems limits their practical usability, highlighting the need for a more comprehensive solution.

3.1.1 DEMERITS OF EXISTING SYSTEM

- Most systems require a stable internet connection, making them unsuitable for rural and remote agricultural areas.
- Cloud-based solutions introduce latency due to data transmission and increase bandwidth consumption.
- Many systems provide generalized treatment suggestions rather than crop-specific and disease-specific recommendations.
- Traditional machine learning models exhibit reduced performance in real-world environments due to variations in lighting, background, and image quality.
- Lack of accurate disease severity estimation in most existing approaches limits effective decision-making.
- Remote sensing techniques are not capable of detecting diseases at the individual leaf level with high precision.
- High operational, deployment, and maintenance costs restrict accessibility for small and marginal farmers.
- Dependence on manual inspection or expert intervention in certain systems reduces efficiency and scalability.
- Lack of integrated decision-support systems combining detection, severity analysis, and recommendation.
- Data privacy and security concerns in cloud-based implementations.
- Requirement of large labeled datasets for training, which may not always be available for all crop diseases.
- Limited user-friendly interfaces in some systems, affecting usability for non-technical users.

3.2 PROPOSED SYSTEM

The proposed system is a Plant Health Monitoring System Using Leaves that utilizes image processing and deep learning techniques for accurate plant disease detection and health assessment. The system is designed to overcome the limitations of existing methods by providing a reliable, efficient, and automated solution.

The system accepts a leaf image as input, either captured through a camera or uploaded by the user. The input image undergoes preprocessing operations such as resizing, normalization, and noise reduction to enhance image quality. Segmentation techniques are applied to isolate the leaf region from the background, ensuring precise analysis.

The processed image is then passed to a Convolutional Neural Network (CNN) model based on a transfer learning approach using EfficientNet-Lite0. The model automatically extracts important features such as color, texture, and patterns and performs multi-class classification to identify the type of plant disease.

In addition to disease detection, the system includes a severity estimation module, which calculates the proportion of infected area on the leaf using pixel-based analysis. A heatmap visualization is generated to highlight affected regions, improving the interpretability of results.

To ensure reliable predictions, the system incorporates a validation mechanism that evaluates image quality and model confidence. The system also functions as a decision support system (DSS) by providing suitable recommendations for fertilizers and pesticides based on the detected disease.

A Graphical User Interface (GUI) is developed to enable easy interaction, allowing users to upload images, view analysis results, and maintain records. Overall, the proposed system provides an automated, accurate, and cost-effective solution for plant health monitoring and disease management.

3.2.1 MERITS OF PROPOSED SYSTEM

- Automated Disease Detection: Identifies plant diseases without manual intervention.
- Deep Learning-Based Classification: Uses CNN with EfficientNet-Lite0 for high accuracy.
- Transfer Learning Approach: Improves performance and reduces training time.
- Multi-Class Classification: Capable of detecting multiple plant diseases.
- Automatic Feature Extraction: Eliminates the need for manual feature engineering.
- Severity Estimation: Calculates infection level using pixel-based analysis.
- Heatmap Visualization: Highlights infected regions for better understanding.
- Validation Mechanism: Ensures reliable predictions using confidence checks.
- Decision Support System: Provides recommendations for fertilizers and pesticides.
- User-Friendly Interface: Enables easy interaction and result visualization.
- Real-Time Processing: Generates quick results for timely action.
- Cost-Effective Solution: Reduces dependency on multiple tools and experts.

CHAPTER 4

REQUIREMENTS SPECIFICATION

4.1 SIMULATION ENVIRONMENT

The simulation environment is established to replicate the operational conditions required for developing and evaluating the proposed plant health monitoring system. It provides a virtual platform where leaf image datasets are processed, deep learning models are trained, and system performance is validated before real-time deployment. This environment enables systematic experimentation with preprocessing techniques, feature extraction methods, and classification models to achieve optimal accuracy and reliability.

The simulation setup supports iterative model training, validation, and testing using structured datasets, ensuring that the system can handle variations in image quality, lighting conditions, and background noise. It also facilitates performance analysis through metrics such as accuracy, loss, and validation scores, allowing fine-tuning of model parameters. By integrating data handling, model training, and evaluation within a single framework, the simulation environment ensures consistency, scalability, and efficiency in system development.

Furthermore, the environment is designed to support GPU-accelerated computation for faster training and reduced processing time. It allows the use of modern deep learning frameworks and libraries, enabling seamless implementation of complex neural network architectures. Overall, the simulation environment plays a crucial role in verifying the functionality, robustness, and effectiveness of the proposed system prior to deployment.

4.2 HARDWARE AND SOFTWARE REQUIREMENTS

4.2.1 Hardware Requirements

The implementation of the proposed system requires a standard computing environment capable of supporting deep learning operations. The hardware specifications are designed to ensure smooth execution during both development and deployment phases.

- **Processor:** Intel Core i5 or higher
- **RAM:** Minimum 8 GB
- **Storage:** 256 GB or above
- **Graphics Processing Unit:** NVIDIA GPU for accelerated training
- **Input Devices:** Camera or image upload interface
- **Output Devices:** Monitor or display unit

4.2.2 Software Requirements

The system is developed using advanced software tools and frameworks that facilitate efficient implementation and model training.

- **Operating System:** Windows / Linux / macOS
- **Programming Language:** Python
- **Frameworks:** TensorFlow and Keras
- **Libraries:** NumPy, OpenCV, Matplotlib
- **Development Tools:** Jupyter Notebook / Visual Studio Code

CHAPTER 5

SYSTEM DESIGN

5.1 ARCHITECTURE DIAGRAM

The proposed Plant Health Monitoring System Using Leaf Image Analysis is designed using a modular and layered architecture to ensure efficient processing, scalability, and accurate disease detection. The system performs a complete workflow starting from image acquisition to disease prediction and recommendation using deep learning techniques.

The architecture integrates multiple components such as image preprocessing, basic segmentation, deep learning-based feature extraction, model training, and prediction modules. Each component is responsible for a specific function, enabling smooth data flow and improved system performance. This structured approach enhances system reliability, maintainability, and allows easy extension of additional features in the future.

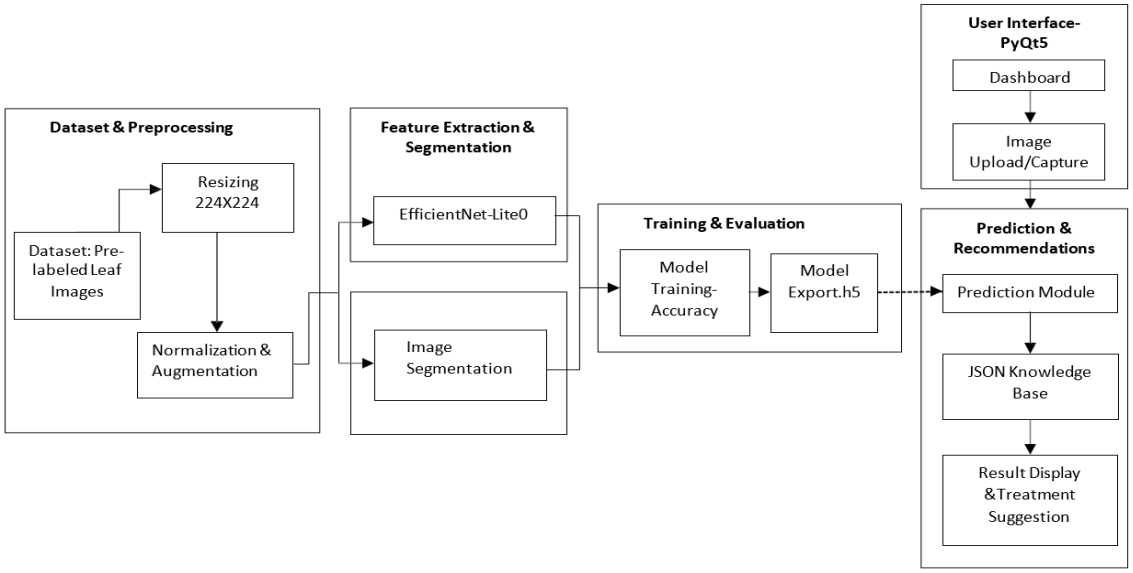


Fig 5.1 ARCHITECTURE DIAGRAM

LAYERED ARCHITECTURE EXPLANATION

The proposed system follows a layered architecture where each layer performs a specific task in the plant disease detection process. This design ensures efficient data processing, modularity, and scalability.

Input Layer (User Interface)

This layer acts as the entry point of the system. The user interacts with the application through a graphical user interface (GUI) developed using PyQt5. The user can upload a leaf image or capture it using a camera. The system validates the input image to ensure it is suitable for further processing.

Preprocessing and Segmentation Layer

In this layer, the input image undergoes preprocessing operations such as resizing to 224×224 pixels and normalization to standardize pixel values. Basic image segmentation techniques are applied to isolate the leaf region from the background, which helps in improving the accuracy of disease detection.

Additionally, data augmentation techniques such as rotation, flipping, and zooming are applied during training to enhance model generalization and robustness.

Deep Learning Feature Extraction Layer

This layer utilizes a Convolutional Neural Network (CNN) based on EfficientNet-Lite0. The system combines basic segmentation with deep learning to improve feature extraction and classification performance. The model automatically extracts important features such as color, texture, and patterns from the input image, eliminating the need for manual feature engineering.

Training and Model Generation Layer

In this layer, the deep learning model is trained using preprocessed dataset

images. The training process is optimized using techniques such as early stopping, learning rate reduction, and model checkpointing to improve accuracy and prevent overfitting.

The trained model is evaluated using performance metrics such as accuracy and is saved in .h5 format for future use in prediction.

Prediction and Recommendation Layer

This layer is responsible for real-time disease detection. The input image is passed through the trained model to classify the type of plant disease. Based on the prediction, the system retrieves appropriate treatment suggestions from a knowledge base stored in JSON format.

This layer acts as a decision support system by providing recommendations such as fertilizers and pesticides to assist users in taking corrective actions.

Output Layer (Result Visualization)

The final layer presents the output to the user in a clear and understandable format. The system displays the predicted disease along with recommended treatments through the graphical interface.

This layer ensures that the results are user-friendly and helps users make informed decisions regarding plant health management.

5.2 EXPECTED OUTCOME

The proposed architecture is designed to achieve high accuracy and efficiency in plant disease detection with enhanced reliability under real-world operating conditions. The system performs precise disease classification and generates relevant and context-aware treatment recommendations, thereby facilitating informed decision-making. Furthermore, it minimizes reliance on expert intervention, improves accessibility for non-technical users, and contributes to effective plant health management and increased agricultural productivity.

CHAPTER 6

SYSTEM IMPLEMENTATION

6.1 PROPOSED APPROACH

The proposed Plant Health Monitoring System is designed to detect plant diseases using leaf image analysis and deep learning techniques. The system follows a structured approach that integrates image processing and machine learning to ensure accurate and efficient disease detection.

The process begins with the user uploading a leaf image through a graphical user interface. The input image is validated and then passed to the preprocessing stage, where operations such as resizing and normalization are performed to maintain consistency.

Basic segmentation techniques are applied to isolate the leaf region from the background, which improves the quality of analysis by removing noise and irrelevant information.

The processed image is then fed into a deep learning model based on a Convolutional Neural Network (CNN) using EfficientNet-Lite0. The model automatically extracts important features such as color, texture, and patterns from the image, eliminating the need for manual feature extraction.

Based on these extracted features, the system classifies the type of plant disease with high accuracy. After classification, the system generates suitable treatment recommendations, including fertilizers and pesticides.

Finally, the results are displayed to the user in a clear and understandable format. This approach ensures a fast, reliable, and user-friendly solution for plant health monitoring.

6.2 SYSTEM MODULES

The system is divided into the following major modules:

- Input Module
- Preprocessing and Segmentation Module
- Deep Learning Module
- Classification Module
- Recommendation Module
- User Interface Module

6.3 MODULE DESCRIPTION

6.3.1 Input Module

The Input Module acts as the entry point of the system. It allows users to upload leaf images through a graphical user interface. The module ensures that the uploaded image is in a valid format (such as JPG or PNG) and suitable for further processing. It simplifies user interaction and provides an easy and efficient way to access the system.

6.3.2 Preprocessing and Segmentation Module

This module prepares the input image for analysis. The uploaded image is resized to a standard dimension (224×224 pixels) and normalized to ensure consistency across all inputs.

Basic segmentation techniques are applied to isolate the leaf region from the background. This helps in reducing noise and improving the accuracy of disease detection by focusing only on the relevant portion of the image.

6.3.3 Deep Learning Module

The Deep Learning Module is the core component of the system. It utilizes a Convolutional Neural Network (CNN) based on EfficientNet-Lite0 for analyzing the processed image.

The model automatically extracts important features such as color, texture, and patterns from the image. This eliminates the need for manual feature engineering and improves the efficiency and accuracy of the system.

6.3.4 Classification Module

The Classification Module is responsible for identifying the type of plant disease based on the features extracted by the deep learning model.

It performs multi-class classification and assigns the input image to one of the predefined disease categories. The module ensures accurate and reliable prediction results.

6.3.5 Recommendation Module

This module provides suitable treatment suggestions based on the detected disease. It recommends appropriate fertilizers and pesticides to help users take corrective actions.

The module acts as a decision support system, assisting users in improving plant health and preventing further damage.

6.3.6 User Interface Module

The User Interface Module manages the interaction between the user and the system. It provides a simple and user-friendly interface for uploading images and viewing results.

The module displays the predicted disease along with recommendations in a clear and organized format, ensuring a smooth and efficient user experience.

CHAPTER 7

RESULTS

7.1 SYSTEM OUTPUT DESCRIPTION

7.1.1 DATASET LOADING AND CLASS IDENTIFICATION

Description:

This output represents the initial stage of the system where the dataset is successfully loaded into the deep learning pipeline. The system reads the training and validation datasets from directory-based storage, where images are organized into separate folders corresponding to different plant disease classes. This structured organization enables automatic labeling of images without manual intervention.

During execution, the system scans the dataset directories and identifies all available classes based on folder names. It then creates an internal mapping between class labels and numerical indices, which is essential for categorical classification. The output generated at this stage displays the total number of classes and the list of detected class names.

This output is significant because it verifies that the dataset has been correctly loaded and structured. Any issues such as missing folders, incorrect directory paths, or improper dataset organization can be identified at this stage itself. It also ensures that the model is aware of all possible categories it needs to classify.

Algorithm

Purpose:

To load dataset from directory structure and identify class labels for classification.

Input:

- Training dataset directory containing class-wise subfolders

- Validation dataset directory
- Leaf images organized based on disease categories
- Directory path configuration provided in the system

Process:

1. Initialize dataset loading mechanism using directory-based data generator.
2. Access the specified training directory path.
3. Traverse through all subdirectories present inside the dataset folder.
4. Identify each subdirectory as a unique class label.
5. Read all image files present inside each class folder.
6. Verify image formats (JPG/JPEG/PNG) for compatibility.
7. Assign numerical indices to each class automatically.
8. Create a mapping of class indices to class names.
9. Repeat the same process for validation dataset directory.
10. Store dataset structure in memory-efficient format.
11. Count total number of classes detected in the dataset.
12. Extract and store class names in a list structure.
13. Display dataset summary including class count and labels.

Output:

- Total number of classes detected in dataset
- List of class names (disease categories)
- Mapping of class labels to numerical indices
- Confirmation of successful dataset loading
- Structured dataset ready for preprocessing and training

7.1.2 IMAGE PREPROCESSING OUTPUT

Description:

This output represents the preprocessing stage where raw input images are transformed into a standardized format suitable for deep learning. Since images collected from real-world agricultural environments may vary in size, resolution, and pixel intensity, preprocessing is essential to ensure uniformity across the dataset.

At this stage, all images are resized to a fixed dimension of 224 × 224 pixels, which matches the input requirements of the EfficientNet model. Additionally, pixel values are normalized from the range [0, 255] to [0, 1]. This normalization improves numerical stability during training and accelerates convergence of the neural network.

The output ensures that all images are consistent in terms of size and scale, thereby preventing computational errors and improving model performance. This stage does not alter the semantic content of the image but prepares it for efficient processing.

Algorithm

Purpose:

To standardize image size and normalize pixel values for deep learning processing.

Input:

- Raw leaf images from dataset
- Images with varying resolutions and pixel intensities
- Predefined target size (224 × 224)

Process:

1. Retrieve raw images from dataset loader.

2. Read image files into memory as numerical arrays.
3. Check image dimensions and formats.
4. Resize each image to fixed dimension (224 × 224 pixels).
5. Ensure all images have 3 color channels (RGB).
6. Convert pixel values from integer format to floating-point format.
7. Apply normalization by dividing pixel values by 255.
8. Scale pixel values to range [0, 1].
9. Validate processed images for consistency.
10. Prepare images for batch-wise processing.
11. Forward preprocessed images to next stage (augmentation/training).

Output:

- Resized images with uniform dimensions
- Normalized pixel values
- Consistent image dataset
- Preprocessed images ready for model input

7.1.3 DATA AUGMENTATION OUTPUT

Description:

This output represents the augmented dataset generated during the training process. Data augmentation is applied to artificially increase the diversity of training data without collecting new images. This is particularly important in plant disease detection, where datasets may be limited or imbalanced.

The system applies multiple transformations such as rotation, shifting, zooming, flipping, and brightness adjustments. These transformations simulate real-world conditions like different camera angles, lighting variations, and leaf orientations.

The output of this stage is not a static dataset but a dynamically generated stream of augmented images. Each training epoch may see slightly different variations of the same image, which helps the model learn more robust and generalized features.

Algorithm

Purpose:

To enhance dataset diversity and improve model generalization using augmentation techniques.

Input:

- Preprocessed training images
- Augmentation parameters (rotation, shift, zoom, etc.)

Process:

1. Initialize augmentation generator with defined parameters.
2. Apply random rotation within specified angle range.
3. Apply horizontal and vertical shifting transformations.
4. Apply zoom-in and zoom-out transformations.
5. Apply shear transformation to distort image geometry.
6. Perform horizontal flipping of images.
7. Adjust brightness levels within defined range.
8. Fill empty pixel regions using nearest neighbor method.
9. Generate augmented images dynamically during training.
10. Ensure each batch contains varied image samples.
11. Continuously feed augmented images to training model.

Output:

- Augmented training image batches
- Increased dataset variability
- Improved model robustness
- Reduced overfitting during training

7.1.4 MODEL ARCHITECTURE OUTPUT

Description:

This output displays the complete architecture of the deep learning model used for plant disease classification. The system utilizes a pre-trained EfficientNet-Lite0 model as a feature extractor, combined with custom classification layers.

The architecture output includes details such as input shape, number of layers, number of parameters, and layer configurations. It clearly shows how the input image flows through different layers, including convolutional layers, dense layers, batch normalization, and dropout layers.

This output is important for verifying that the model has been correctly constructed and configured before training. It also provides insights into the complexity and capacity of the model.

Algorithm

Purpose:

To define and visualize deep learning model structure.

Input:

- Input image shape (224 × 224 × 3)
- Pretrained EfficientNet model
- Custom classification layer configuration

Process:

1. Define input layer with specified shape.
2. Load pre-trained EfficientNet feature extractor.
3. Freeze feature extractor weights (non-trainable).
4. Pass input through feature extraction layer.
5. Add fully connected dense layer (256 neurons).
6. Apply ReLU activation function.
7. Apply batch normalization to stabilize learning.
8. Add dropout layer to prevent overfitting.
9. Add second dense layer (128 neurons).
10. Apply batch normalization and dropout again.
11. Define output layer with softmax activation.
12. Match output neurons with number of classes.
13. Compile complete model architecture.
14. Generate and display model summary.

Output:

- Model summary showing layers and parameters
- Total trainable and non-trainable parameters
- Structured architecture for training
- Verified deep learning model ready for execution

7.1.5 USER LOGIN PAGE OUTPUT

Description:

The User Login Page represents the initial interaction point between the user and

the system. This interface is designed to authenticate users before granting access to the plant disease detection functionalities. The system ensures that only valid users can proceed further, thereby maintaining data integrity and controlled access.

The login page consists of input fields for username and password, along with validation mechanisms to verify user credentials. The system performs authentication by comparing entered credentials with stored user data. Upon successful validation, the user is redirected to the main dashboard. In case of invalid credentials, an error message is displayed without allowing further access.

This output ensures secure and personalized usage of the application while preventing unauthorized access.

Functional Output Details

Input:

- Username entered by user
- Password entered by user

System Processing:

1. Accepts user credentials through input fields
2. Validates input format (non-empty fields)
3. Compares entered credentials with stored data
4. Verifies authentication status
5. Handles incorrect login attempts
6. Redirects valid users to dashboard
7. Restricts access for invalid users

Output Displayed:

- Successful login message
- Redirection to main interface
- Error message for invalid login

7.1.6 DISEASE PREDICTION OUTPUT

Description:

This output represents the core functionality of the system, where the trained deep learning model predicts the plant disease from a given leaf image. The user uploads or captures an image, which is processed and analyzed by the model.

The system extracts learned features from the image and computes prediction probabilities for each class. The class with the highest probability is selected as the final prediction. The result is displayed clearly to the user along with confidence level.

This output provides real-time disease identification, enabling users to take appropriate agricultural actions.

Functional Output Details

Input:

- Leaf image uploaded by user

System Processing:

1. Accepts input image from user interface
2. Validates image format (JPG/JPEG/PNG)
3. Resizes image to required dimensions
4. Normalizes pixel values
5. Passes image through trained model

6. Extracts prediction probabilities
7. Identifies highest probability class
8. Maps class index to disease label

Output Displayed:

- Predicted disease name
- Confidence score (%)
- Classification result

7.1.7 INVALID IMAGE HANDLING OUTPUT

Description:

This output handles cases where the input image does not meet the required criteria for prediction. The system ensures robustness by validating the input image before processing.

If the uploaded image is corrupted, unsupported, or does not represent a leaf image, the system prevents further processing and displays an appropriate error message. This avoids incorrect predictions and improves system reliability.

Functional Output Details

Input:

- Uploaded image (invalid or unsupported)

System Processing:

1. Checks file format compatibility
2. Verifies image readability
3. Detects corrupted or empty files
4. Validates image dimensions

5. Identifies non-leaf or irrelevant inputs (basic validation)
6. Stops processing if validation fails

Output Displayed:

- Error message for invalid image
- Instruction to upload valid leaf image
- No prediction result generated

7.1.8 FEATURE EXTRACTION OUTPUT

Description:

This output represents the internal feature extraction performed by the deep learning model. Although not directly visible to the user, the system processes and identifies important visual features such as color variations, texture patterns, and disease spots from the leaf image.

The EfficientNet-based model automatically extracts high-level features that distinguish healthy and diseased leaves. These extracted features are used for classification and contribute to prediction accuracy.

For visualization purposes (if enabled), feature maps or highlighted regions can be displayed to indicate important areas influencing the prediction.

Functional Output Details

Input:

- Preprocessed leaf image

System Processing:

1. Passes image through convolutional layers
2. Extracts low-level features (edges, shapes)
3. Extracts mid-level features (textures, patterns)

4. Extracts high-level features (disease regions)
5. Reduces feature dimensionality
6. Converts features into numerical representation
7. Feeds extracted features to classification layers

Output Displayed:

- Extracted feature representation (internal)
- Optional visualization of important regions
- Feature vector used for classification

7.1.9 DISEASE SEVERITY COMPARISON OUTPUT

Description:

This output provides an estimation of the severity level of the detected disease. The system analyzes the affected regions in the leaf image and compares them with predefined patterns or thresholds.

Severity is categorized into levels such as Low, Medium, and High based on the extent of damage observed. This helps users understand how critical the disease condition is and prioritize treatment accordingly.

Functional Output Details

Input:

- Predicted disease class
- Extracted feature patterns

System Processing:

1. Analyzes affected regions in image
2. Compares intensity of disease features
3. Measures spread of infection patterns

4. Applies predefined severity thresholds
5. Classifies severity level
6. Associates severity with prediction result

Output Displayed:

- Severity level (Low / Medium / High)
- Comparative indication of damage
- Supporting information for decision-making

7.1.10 DIAGNOSIS HISTORY OUTPUT

Description:

This output maintains a record of all previous predictions made by the system. It allows users to track historical diagnosis results and analyze patterns over time.

The system stores details such as input image, predicted disease, confidence score, and timestamp. This feature enhances usability by enabling users to review past analyses and monitor plant health progression.

Functional Output Details

Input:

- Prediction results generated by system

System Processing:

1. Stores prediction details after each analysis
2. Records disease name and confidence score
3. Saves timestamp of prediction
4. Maintains structured history log
5. Retrieves stored records when requested

6. Displays history in user interface

Output Displayed:

- List of previous predictions
- Disease names and results
- Date and time of analysis
- Historical tracking of plant health

7.1.11 RECOMMENDATION OUTPUT

The Recommendation Output Module generates treatment suggestions based on the predicted plant disease. After classification, the system maps the predicted disease label to a predefined dataset containing suitable fertilizers, pesticides, and preventive measures. The recommendations are retrieved from a structured JSON file and presented to the user in a clear and organized format. This module enables users to take appropriate corrective actions by converting prediction results into practical agricultural guidance.

Algorithm

Purpose:

To generate appropriate treatment recommendations based on predicted plant disease

Input:

- Predicted disease class label
- Recommendation dataset (JSON file)

Process:

1. Receive the predicted disease label from the classification module
2. Load the recommendation dataset stored in JSON format
3. Search for the matching disease entry in the dataset

4. Retrieve associated details such as:
    - Recommended pesticides
    - Suggested fertilizers
    - Organic or alternative treatments
    - Preventive measures
5. Verify the availability of recommendation data for the given disease
6. Structure the retrieved data into a readable format
7. Send formatted output to the user interface for display

Output:

- Disease-specific treatment recommendations
- Suggested fertilizers and pesticides
- Preventive guidelines
- Structured output displayed to the user

7.2 SCREENSHOTS

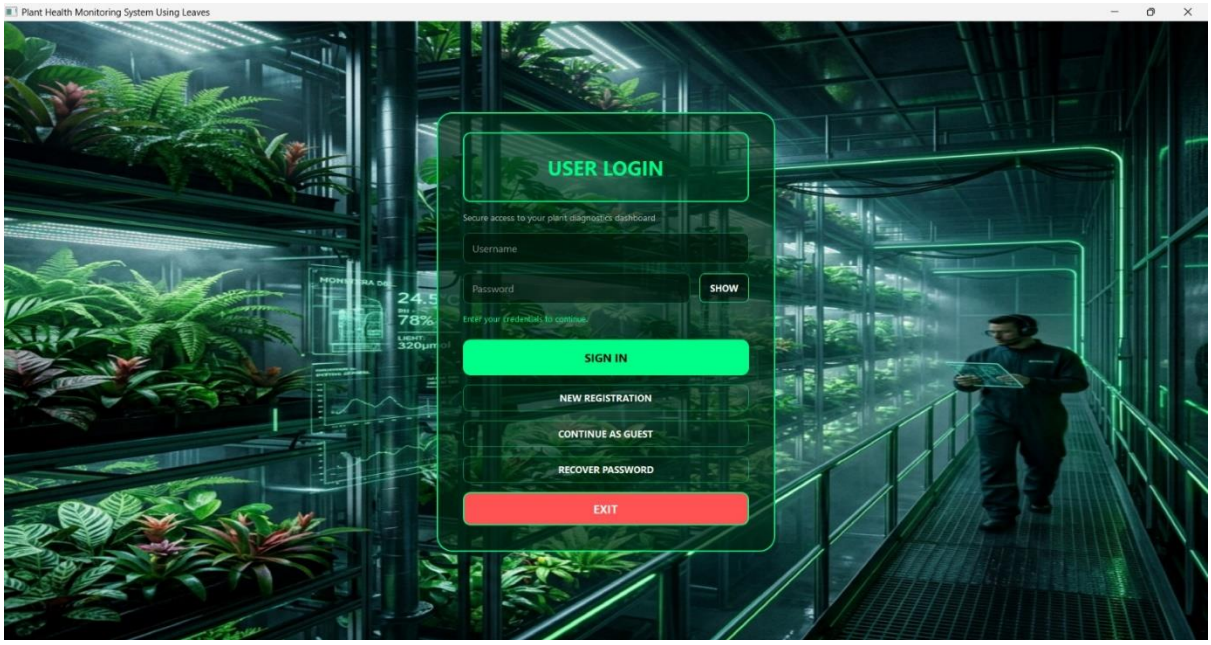


Fig 7.2.1 User Login

Figure 7.2.1 illustrates the user login interface where users enter their credentials to access the system. It ensures authentication and controlled access to the application features.



Fig 7.2.2 Disease Prediction

Figure 7.2.2 shows the disease prediction output where the system analyzes the uploaded leaf image and displays the identified plant disease along with the corresponding confidence score.

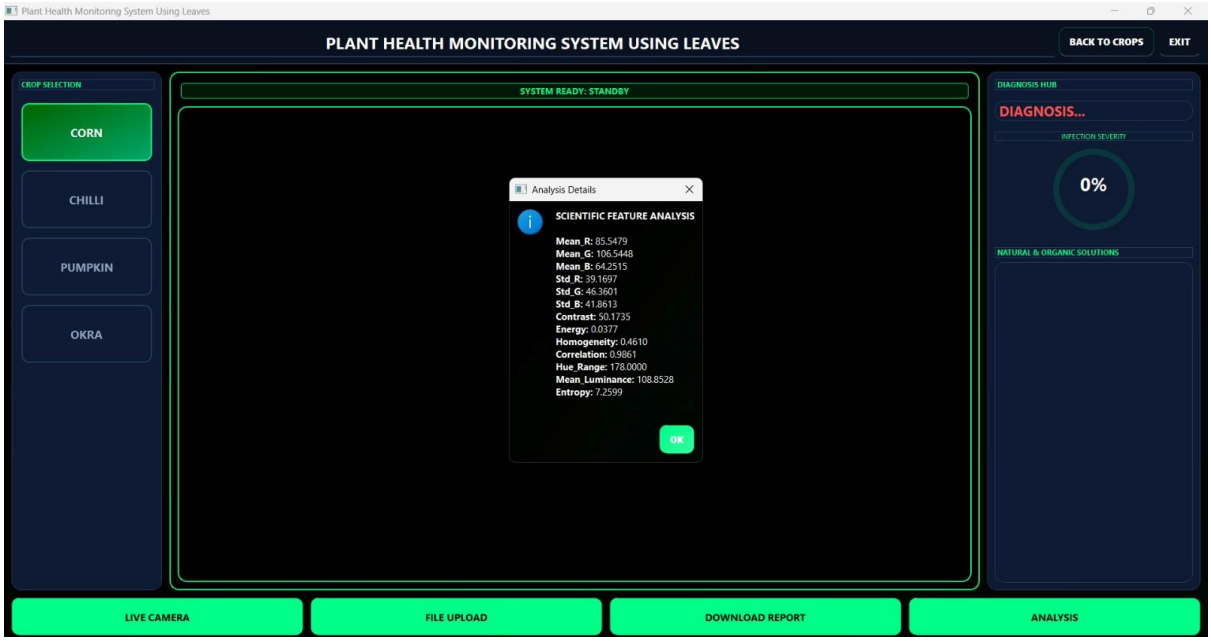


Fig 7.2.3 Extracted Features

Figure 7.2.3 represents the feature extraction process, highlighting the important regions and patterns in the leaf image that are used by the model for disease classification.

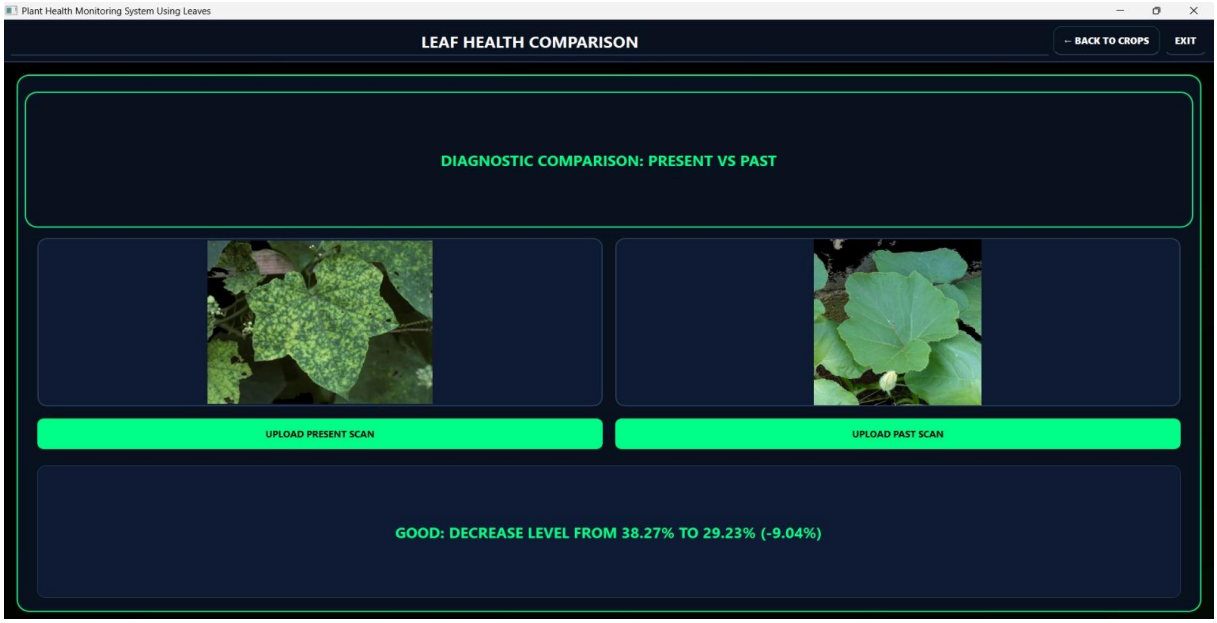


Fig 7.2.4 Severity Comparison

Figure 7.2.4 displays the severity analysis of the detected disease, indicating the level of infection based on the affected area in the leaf.

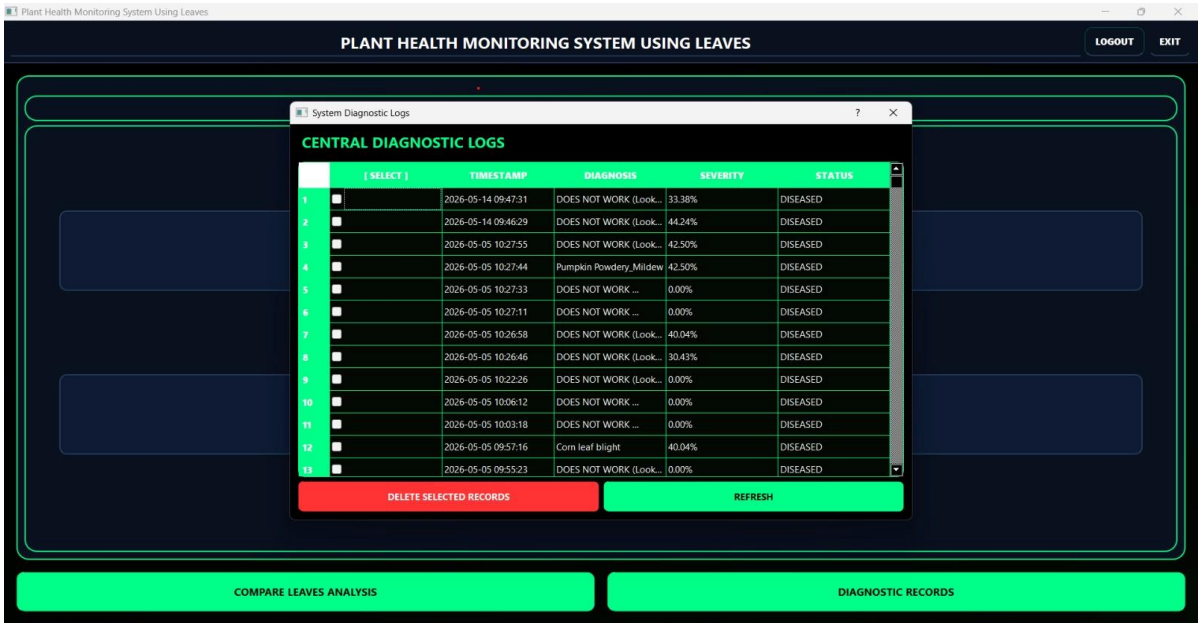


Fig 7.2.5 Diagnosis History

Figure 7.2.5 illustrates the diagnosis history, showing previously analyzed results including predicted diseases, confidence scores, and timestamps for tracking plant health over time.

CHAPTER 8

CONCLUSION AND FUTURE ENHANCEMENT

8.1 CONCLUSION

The proposed Plant Health Monitoring System using Leaf Image Analysis provides an effective and automated solution for detecting plant diseases. The system utilizes image processing techniques and deep learning models to analyze leaf images and identify diseases with high accuracy.

By integrating preprocessing, segmentation, and a Convolutional Neural Network (CNN) based on EfficientNet-Lite0, the system is capable of extracting important features such as color, texture, and patterns from leaf images. This enables accurate classification of plant diseases without the need for manual feature extraction.

The system also includes a recommendation module that suggests suitable fertilizers and pesticides based on the detected disease. This helps users take timely corrective actions and improve crop health. The user-friendly interface ensures easy interaction, making the system accessible even to non-technical users.

Overall, the developed system is fast, reliable, and cost-effective. It reduces dependency on manual inspection and expert knowledge, thereby supporting modern agricultural practices. The project successfully demonstrates how deep learning can be applied to real-world problems in agriculture to improve productivity and sustainability.

8.2 FUTURE ENHANCEMENT

Although the system performs effectively, there are several areas where further improvements can be made to enhance its performance and usability:

- **Expansion of Dataset**

  Increasing the size and diversity of the dataset can improve model accuracy and enable detection of a wider range of plant diseases.

- **Real-Time Disease Detection**

  Integration with mobile devices or cameras can enable real-time detection directly in agricultural fields.

- **Integration with IoT Systems**

  Sensors can be used to monitor environmental conditions such as temperature and humidity, providing better insights into plant health.

- **Multi-Language Support**

  Adding regional language support can make the system more accessible to farmers in different areas.

- **Cloud-Based Deployment**

  Deploying the system on cloud platforms can allow remote access and large-scale usage.

- **Disease Severity Prediction Enhancement**

  More advanced techniques can be used to improve the accuracy of severity estimation and visualization.

- **Automated Alert System**

  Notifications can be sent to users when a disease is detected, enabling quick action.

REFERENCES

[1] S. P. Mohanty, D. P. Hughes, and M. Salathé, “Using deep learning for image-based plant disease detection,” *Frontiers in Plant Science*, vol. 7, p. 1419, 2016.

[2] K. P. Ferentinos, “Deep learning models for plant disease detection and diagnosis,” *Computers and Electronics in Agriculture*, vol. 145, pp. 311–318, 2018.

[3] M. Tan and Q. V. Le, “EfficientNet: Rethinking model scaling for convolutional neural networks,” in *Proceedings of the International Conference on Machine Learning (ICML)*, 2019, pp. 6105–6114.

[4] C. Shorten and T. M. Khoshgoftaar, “A survey on image data augmentation for deep learning,” *Big Data*, vol. 6, no. 1, pp. 60–76, 2019.

[5] R. R. Selvaraju, M. Cogswell, A. Das, R. Vedantam, D. Parikh, and D. Batra, “Grad-CAM: Visual explanations from deep networks via gradient-based localization,” in *Proceedings of the IEEE International Conference on Computer Vision (ICCV)*, 2017, pp. 618–626.

[6] A. Too, L. Yujian, S. Njuki, and L. Yingchun, “A comparative study of fine-tuning deep learning models for plant disease identification,” *Computers and Electronics in Agriculture*, vol. 161, pp. 272–279, 2019.

[7] S. Sladojevic, M. Arsenovic, A. Anderla, D. Culibrk, and D. Stefanovic, “Deep neural networks based recognition of plant diseases by leaf image classification,” *Computational Intelligence and Neuroscience*, vol. 2016, pp. 1–11, 2016.

[8] P. B. P. Reddy, M. S. Kumar, and B. S. Reddy, “Plant leaf disease detection using deep learning and convolutional neural networks,” *International Journal of Engineering and Advanced Technology*, vol. 8, no. 6, pp. 1–5, 2019.